

Problem Solving Using Python (CSE1012)

Dr. Abdul Kayom, Assistant Professor, SENSE

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Number Conversions

- Write Prog to convert one number system to another
 - Decimal to binary
 - Binary to decimal
 - etc.

Strings

- Creating and Accessing Strings
- Operations on Strings
- Indexing
- Slicing
- String Manipulations
- Pre-defined functions on Strings

What is a Random Number?

- Random number does NOT mean a different number every time.
- Random means something that can not be predicted logically.

Pseudo Random and True Random

- Computers work on programs, and programs are definitive set of instructions.
- So it means there must be some algorithm to generate a random number as well.
- If there is a program to generate random number it can be predicted, thus it is not truly random.

Pseudo Random and True Random

- Random numbers generated through a generation algorithm are called ***pseudo random***.
- Can we make truly random numbers?

Pseudo Random and True Random

- Can we make truly random numbers?
- Yes. In order to generate a truly random number on our computers we need to get the random data from some outside source.
- This outside source is generally our keystrokes, mouse movements, data on network etc.

Pseudo Random and True Random

- We do not need truly random numbers, unless it is related to security (e.g. encryption keys) or the basis of application is the randomness (e.g. Digital roulette wheels).

Random Numbers using Numpy

Generate a random integer from 0 to 100:

```
import numpy as np
```

```
x1 = np.random.randint(100)
```

```
print(x1)
```

Random Numbers using Numpy

#Generate a random float from 0 to 1:

```
import numpy as np
```

```
x2 = np.random.rand()
```

```
print(x2)
```

Random Numbers using Numpy

The randint() method takes a size parameter where you can
specify the shape of an array.

Example:

Generate a 1-D array containing 5 random integers from 0 to 100:

```
import numpy as np
x3 = np.random.randint(100, size=(5))
print("Random Array of numbers")
print(x3)
```

Random Numbers using Numpy

```
# Generate a 2-D array with 3 rows, each row containing 5  
# random integers from 0 to 100:
```

```
import numpy as np  
x4 = np.random.randint(100, size=(3, 5))  
print("Random 2-D Array of numbers")  
print(x4)
```

Random Numbers using Numpy

Generate a 1-D array containing 5 random floats:

```
import numpy as np  
x5 = np.random.rand(5)  
print(x5)
```

Random Numbers using Numpy

#Return one of the values in an array:

```
import numpy as np  
x6 = np.random.choice([3, 5, 7, 9])  
print(x6)
```

Random Numbers using Numpy

```
# Generate a 2-D array that consists of the values in the array  
# parameter (3, 5, 7, and 9):
```

```
import numpy as np  
x7 = np.random.choice([3, 5, 7, 9], size=(3, 5))  
print(x7)
```

Thank You

Dr. Abdul Kayom



Asst. Prof, (SENSE)



kayomabdul@vitap.ac.in



8474860025